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Spatio-temporal measurement of one-dimensional surface potential distributions using a surface voltmeter and a slit plate

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Abstract

Fine parallel electrodes with an electrode pitch ranging from sub-mm to mm have been utilized in electrostatic devices. Evaluating the surface potential distributions of those electrodes are essential in characterizing the electrostatic devices. However, the spatial resolution of commercially available surface voltmeters are insufficient for the evaluations. Targeting at one-dimensional distributions, this work improves the spatial resolution of a voltmeter using a slit plate. A slit plate narrows the field-of-view and realizes a higher resolution in measurement of one-dimensional potential distributions created by parallel electrodes. Field simulations showed that a thin and narrow slit can improve the spatial resolution. Experiments using a slit with a thickness of 0.02 mm and a slit opening of 0.2 mm confirmed the improvement in measurement of a potential distribution produced by parallel electrodes with an electrode pitch of 1 mm. The method was extended for visualizing time variations of surface potential distributions, in which the slit plate was oscillated for scanning. Using the extended method, a potential distribution was scanned at 40 frames per second. The method was applied to evaluation of surface potential distributions of an electrostatic film motor that has three-phase electrodes with an electrode pitch of 0.2 mm. The result of the potential measurement coincided with the force measurement in terms of their break frequencies.

Keywords: surface potential distribution, non-contact surface voltmeter, spatial resolution, high resolution, electrostatic motors, parallel electrodes

(Some figures may appear in colour only in the online journal)

1. Introduction

Recently, the use of electrostatic devices has been expanding in the fields of robotics and human computer interaction (HCI). In robotics, for example, electro-adhesion has been introduced for wall climbing robots [1–4], as well as for robotic inter-connections [5] and robotic grippers [6–8]. High-power electrostatic actuators are another interesting category of electrostatic devices in the field of robotics [9–16]. In HCI, electro-vibration has been gathering attentions, as it can render tactile feeling on a touch panel [17–21].

Those electrostatic devices are typically realized using thin sheets with embedded electrodes. The embedded electrodes

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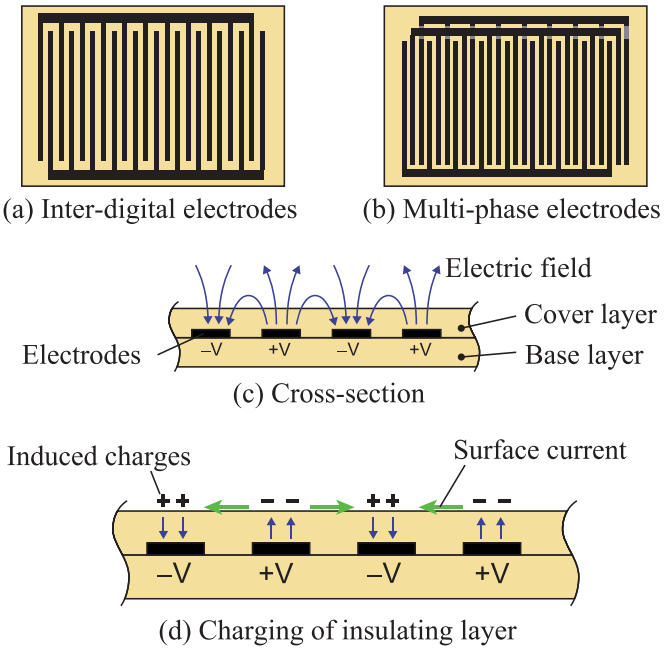


Figure 1. Multi-phase electrodes used in electrostatic devices; (a) inter-digital; (b) multi-phase (three-phase); (c) cross-sectional view and created electric field; (d) charges by surface current alter the electric field above the surface.

can be either planar or multi-phase electrodes. For example, in electro-adhesion devices, while planar electrodes can be utilized to adhere to conductive walls, inter-digital electrodes, which are two-phase parallel electrodes, are often utilized as it realizes adhesion to a wide range of materials including dielectrics. In electrostatic motors, multi-phase parallel or radial electrodes are utilized to create traveling electric fields. Those inter-digital or multi-phase electrodes typically have electrode pitches in the range of sub-millimeters or millimeters.

Those electrodes are normally covered with insulating materials to avoid discharge or short-circuit, as illustrated in figure 1. The insulating materials, however, are not really insulating; they have finite volume resistivity. Because of the finite resistivity, electric current can flow through the body. Their surfaces may have different behaviors from the internal body, since the electric properties of the surfaces can be easily affected by the environment [22, 23]. By absorbing gasses or moisture, for example, the surface resistivity can decrease that allows surface current to flow much more easily than the volume current. Whether surface current or volume current, the current will create a charge distribution on the surface of the insulating layer. Alternatively, dielectric polarization within the insulating layer can also create a charge distribution on the surface. If the surface of the insulating layer is charged, it will change the electric field created by the multi-phase electrodes. Such a change in electric field would affect the performance of the electrostatic devices. In fact, several papers reported that output force changed under constant voltages [17, 24–28], which would be due to the charging.

Since the surface characteristics depend on the environment, it is considerably difficult to estimate the electric behavior theoretically. Therefore, experimental characterization of the behavior of the insulating layer is imperative for characterizing the performance of the electrostatic devices. One practical way to investigate the behavior is to measure the surface potential. If the insulating layer behaves as a perfect insulator, the surface potential of the insulating layer would be almost identical to the electric potential of the electrodes under the layer. If, however, the surface is charged, the surface potential will differ from the potential of the electrodes. Therefore, by measuring the surface potential and its variation along time, we can tell whether and how the insulating layer has been charged.

Surface potentials can be measured using commercially available non-contact surface voltmeters [29, 30]. However, the spatial resolution of a typical non-contact surface voltmeter is several millimeters, which is larger than the electrode pitch of the inter-digital or multi-phase electrodes. As a result, visualizing the surface potential distributions of those electrodes is considerably difficult with typical voltmeters. Efforts have been done to improve the resolution of surface voltmeters. For example, in [31], miniaturized probes are used in an array to realize high-resolution scanning. However, the proposed system requires specially designed probes. Therefore, visualizing the surface potential distribution of those electrostatic devices is still a challenging issue.

There exist other measurement methods that realize finer spatial resolutions. For example, scanning probe microscopy realizes measurements of surface potential distributions with nanometer to sub-micrometer accuracy [32–35]. However, they have limitations in their field-of-view and scanning speed and thus are not suitable for typical electrostatic devices. Another measurement method is utilizing Pockels effect [36–39]. By combining a Pockels cell with a laser system, surface potential can be measured with a considerably high spatio-temporal resolution. Pockels effect might be one solution for investigating the electrostatic devices; however, it requires a considerably complicated setup that is not widely available.

With that in mind, this work proposes a much simpler and cost-effective method to measure the surface potential distributions of the parallel electrodes of electrostatic devices. The method takes advantage of the characteristics of the potential distributions of the parallel electrodes; their distributions are one-dimensional and are uniform along the longitudinal direction of the electrodes. To effectively measure the one-dimensional distributions, the proposed method combines a commercially available surface voltmeter and a slit plate. The slit functions as an aperture to narrow the field of view, which improves the spatial resolution.

The rest of the paper is structured as follows. In the next section, the paper proposes a measurement concept utilizing a slit. The section characterizes the concept through electric field simulation. Then, experiments using a commercially available voltmeter verifies that a slit can improve the spatial resolution of the surface potential measurement. In section 3, the method is extended to visualize a time variation of a surface potential distribution. The slit will be oscillated over the target surface using a linear motor. Then, a potential distribution will

be scanned every half-cycle of the oscillation. In section 4, the proposed method is applied to an electrostatic film motor. The section first shows that the resolution of a commercially available surface voltmeter is not sufficient to evaluate the surface potential of the electrostatic motor. Then, two different motor films, with one being deteriorated in terms of insulating performance, are measured using the proposed method. The result shows that the surface potential distribution decays over time with the deteriorated film, whereas the distribution is kept constant for non-deteriorated one. The result is compared with the result of thrust force measurement of the motor, which indicates that the spatio-temporal measurement of surface potential is effective in characterizing the performance of the motor. Finally, section 5 provides conclusions.

2. Measurement of spatial potential distributions through a slit plate

2.1. Non-contact surface voltmeter

A non-contact surface voltmeter is a sensor that measures surface potential using electrostatic induction. There are several different surface voltmeters in the market, and their operating principles and structures slightly differ from each other. In this paper, we utilize a commercially available surface voltmeter (Model 344 with a probe 6000B-8, TREK, Inc.).

Regardless of the types, the spatial resolutions of commercially available surface voltmeters are typically in the range of millimeters. The resolutions are not fine enough to evaluate the surface potential of fine parallel electrodes, as will be demonstrated later in section 4. To improve the measurement resolution, this work proposes to utilize a slit plate, as shown in figure 2. Here, the slit is expected to function as an aperture to narrow the field of view of the probe of the voltmeter. By fixing the potential of the slit plate at a constant value, the voltmeter would only respond to the change of the electric field that leaks from the slit. In this work, the slit potential is fixed to the ground potential, which is 0 V.

Introduction of a slit plate raises a concern on the lowering of measurement sensitivity. Since the probe measures the average potential over the original field of view, the introduction of the grounded plate will considerably lower the measured potential. If we have to measure two-dimensional distributions, a pin hole must be utilized instead of a slit, to improve the resolution. Since the area of a pin hole is too small, it would be impossible to obtain meaningful measurement results. In this work, however, the target is a one-dimensional pattern that allows us to utilize a one-dimensional slit. A slit can have much larger area than a pin hole and would provide meaningful results, maintaining a certain sensitivity.

2.2. Electrostatic field simulation

In the proposed concept, the electric field from the target surface must leak from the slit to be detected by the probe. To guarantee it, the aspect ratio of the slit needs to be optimized. For that purpose, electric field behavior around a slit

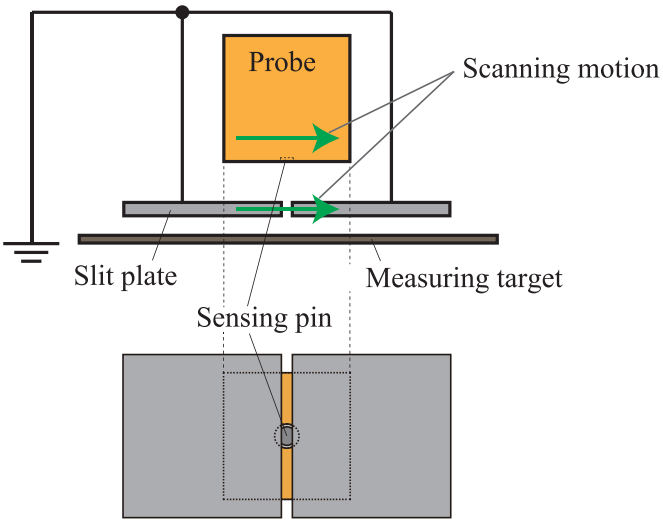


Figure 2. The proposed measurement concept. A grounded slit plate is arranged beneath a probe of a surface voltmeter to narrow the field of view. Then, the probe and the slit together scan the target surface to measure a one-dimensional potential distribution.

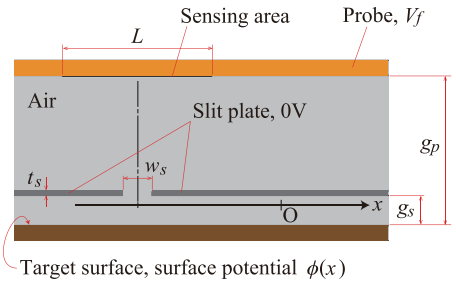


Figure 3. Simulation model. For scanning the target surface, the sensing area and the slit are shifted in the lateral direction.

was simulated using a finite element method (FEM) software (COMSOL Multiphysics 5.5).

Figure 3 shows an example of the two-dimensional simulation models. The model contains a target surface, a slit, and a probe. The target surface is an insulator with a potential distribution. The slit was a conductive body with a potential of 0 V, and had an opening of w_s, which were changed among 0.2, 0.4, and 0.6 mm. The height of the bottom surface from the target, g_s, was fixed to 0.2 mm. The thickness, t_s, was changed from 0.01 to 0.4 mm. The probe was modeled as a single planar electrode and was arranged at a height, g_p, of 1 mm. In the actual probe, a small conductive pin is arranged within a conductive surface of the probe, which is utilized to sense the electric field. In the simulation, the pin was not explicitly modeled but was integrated with the probe body. Then, the central area of the probe, with a length of L, was regarded as the sensing area that provides the same function as the pin in the actual probe.

The simulation was done in the following manner, considering the working principle of the voltmeter. The voltmeter used in this work is a feedback type, in which the potential of the probe body is controlled such that the field beneath the pin becomes zero [40, 41]. Then, the resulting probe potential is used as the output of the voltmeter. To resemble this in the

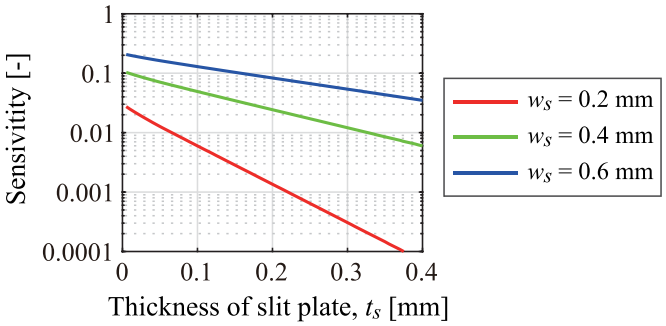


Figure 4. Simulated sensitivity of the proposed method for various slit geometries. Sensitivity decreases rapidly as the slit plate becomes thicker.

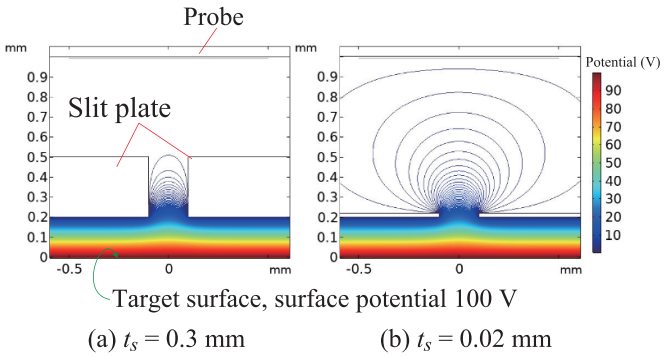


Figure 5. Electric field distributions for thick and thin slit plates. A thicker slit shields the electric field from the target surface.

simulation, we estimated the potential of the probe that creates zero electric field beneath the sensing area.

First, we set 0 V to the probe and calculated the average electric field, $\bar{E}_0$, beneath the sensing area of the probe. Then, we set a enough large potential, V_f, that alters the polarity of the field from the above case. Then, we calculated the average field, $\bar{E}_\mathrm{f}$, for the voltage. Assuming a linear relation, the output potential, where the field becomes zero, was approximated as,

$$V_\mathrm{out} = \frac{\bar{E}_0}{\bar{E}_0 - \bar{E}_\mathrm{f}} V_\mathrm{f}. \tag{1}$$

In the first set of simulations, we fixed the horizontal location of the slit and the probe at the center and estimated the sensitivity. The length L of the sensing area was explored and was set to 0.8 mm, as it provided similar tendencies with the real probe in a pilot evaluation. The target surface was given a uniform potential of 100 V, and the output of the voltmeter was calculated. The ratio of the output voltage against the input represents the sensitivity. Figure 4 shows the results for three different slit widths. The plot shows that slit should be wider and/or thinner to provide higher sensitivity. Especially, the thickness has a considerable impact on the sensitivity. Figure 5 shows some simulation examples that illustrates the electric fields for two different thicknesses. With a thick slit, the electric field from the target is absorbed by the side walls of the slit and does not appear above the slit. On the other hand, with a thin slit plate, the field appears above the slit, which allows the probe to respond to the target potential.

Next, we evaluated the spatial resolution. The target surface was given a potential distribution expressed as follows.

$$\phi(x) = \begin{cases} 0 & (|x| \geqslant s/2) \\ 100 & (|x| < s/2) \end{cases} \tag{2}$$

The width, s, was set to 0.001 mm. Due to the small s, the resulting function can be considered as an impulse function. Then, the probe and the slit plate were moved together, as conceptually shown using green arrows in figure 2 and the output of the probe was calculated at each position. Figure 6 shows the results for three different slit widths. Since the potential on the target is an impulse, the results in figure 6 expresses point spread functions for the corresponding slit widths. The half-value width of each point spread function is regarded as the

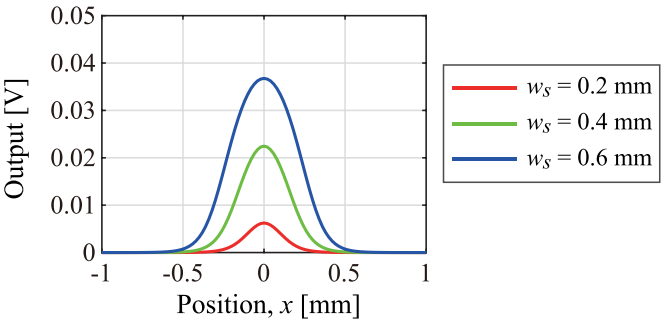


Figure 6. Simulated results of the potential measurements for $g_\mathrm{p} = 1$ mm, $g_\mathrm{s} = 0.2$ mm, $t_\mathrm{s} = 0.02$ mm. As the target potential is an impulse, the results are equivalent with the point spread functions.

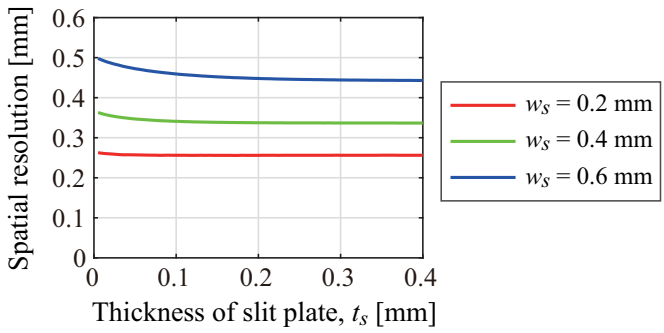


Figure 7. Spatial resolutions obtained as the half-value width of the point spread functions in figure 6.

spatial resolution, which is plotted in figure 7. This plot shows that the resolution becomes worse as the slit becomes wider, whereas the resolution is less affected by the thickness of the slit plate.

As shown in figure 4, the output becomes larger as t_s becomes smaller. On the other hand, figure 7 shows that the resolution is almost independent of t_s. These results are suggesting that the thickness should be as thin as possible. Regarding w_s, smaller w_s results in better resolution, but it also leads to lower sensitivity. Since the sensitivity decreases logarithmically, while the resolution decreases rather linearly, it would be optimal to take the widest w_s that can satisfy the requirement for resolution. Although it is not addressed in this

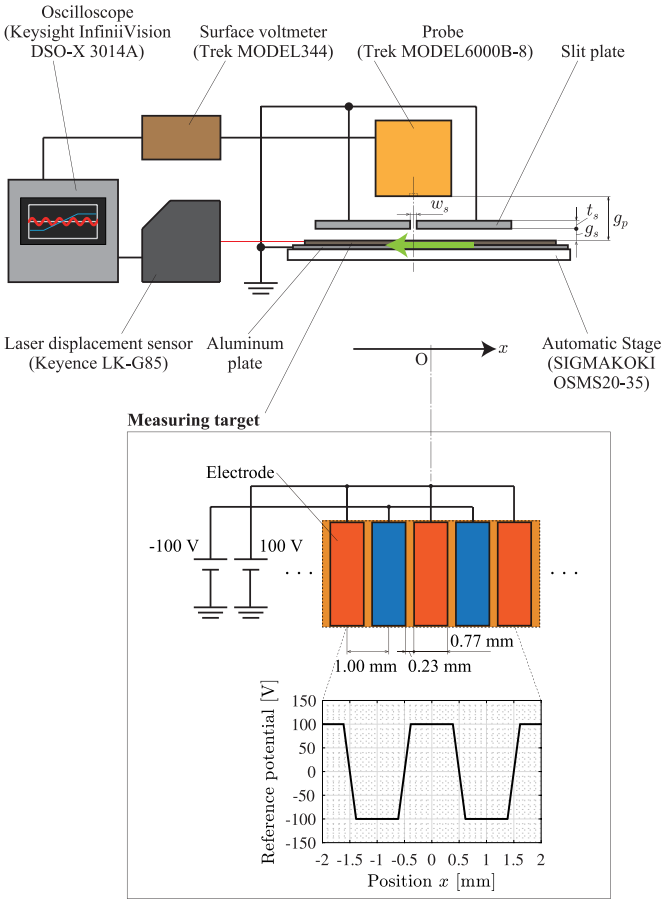


Figure 8. Experimental setup. The target was arranged on a linear stage that moved the target to realize scanning (as shown using a green arrow). The measured target is parallel electrodes connected into two-phase.

paper, when measuring a decaying and spreading distribution, it might be an interesting strategy to change w_s during the measurement to follow the spread of the distribution to maximize the sensitivity.

2.3. Experiment

To verify the feasibility of the concept, surface potential distributions were measured for a film substrate. The film substrate used for this measurement was an flexible printed circuit (FPC) with parallel electrodes. The base and cover films are made of polyimide and the electrodes are made of copper. Its electrode pattern is shown in figure 8 with the experimental setup. A slit plate was arranged below the probe of the surface voltmeter, both of which were fixed to the setup. The target film was arranged on an automatic linear stage (OSMS20-35, SIGMAKOKI Co., Ltd), such that it can move laterally to allow scanning. Note that, whereas the set of the probe and the slit was moved in the simulation, the target was moved in the experiment to make the system configuration simpler. The electrodes in the film were connected in two-phase and were applied ± 100 V.

Three different slits were made using two shim tapes (SFGSM0.02 and SFGSM0.3, Misumi) that have thicknesses,

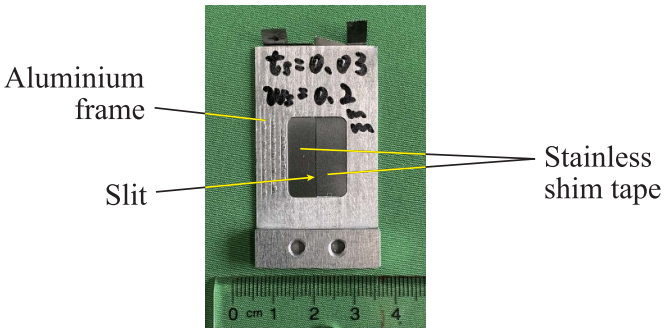


Figure 9. A slit used in the experiments. Two shim tapes were arranged with a narrow opening and were attached to an aluminium frame. The narrow opening functions as the slit.

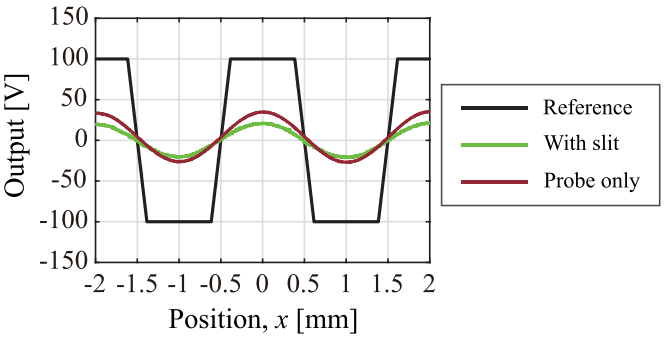


Figure 10. Measurement results using the probe alone (red line) and with a wide slit having $w_s = 1$ mm and $t_s = 0.3$ mm (green line). The height of the slit g_s was set to 0.2 mm. The black line indicates the estimated true value.

t_s, of 0.02 and 0.3 mm. Each of the slits was made of two stainless shim tapes attached to an aluminium frame that was electrically grounded, as shown in figure 9. One slit had an opening, w_s, of 1 mm with a thickness of 0.3 mm, that was used for the measurement in figure 10. The other two slits had openings of 0.2 mm. The height of the probe, g_p, from the measured surface was set to 1 mm. By moving the automatic stage at a constant speed (1 mm s^{-1}), as indicated by the green arrow in figure 8, the surface distribution of the surface potential was measured.

First, as a reference, the measurement result for the wide slit opening ($t_s = 0.3$ mm and $w_s = 1$ mm), as well as the result for the probe alone (without any slit) are shown in figure 10. The black line indicates the estimated true value, whereas the red line shows the result obtained by the probe alone. For the estimated true value, the potential between the electrodes are linearly interpolated. Due to the limited resolution of the probe, the measured distribution looks almost sinusoidal, lacking high spatial frequency components. The green line shows the result with the wide slit opening. The measurement was carried out to confirm that the existence of the grounded plate does not disrupt the measurement of the voltmeter. The obtained distribution is mostly the same as that of the probe alone, indicating that the measurement can be achieved in a stable manner. However, the wide slit does not contribute to the improvement of the resolution.

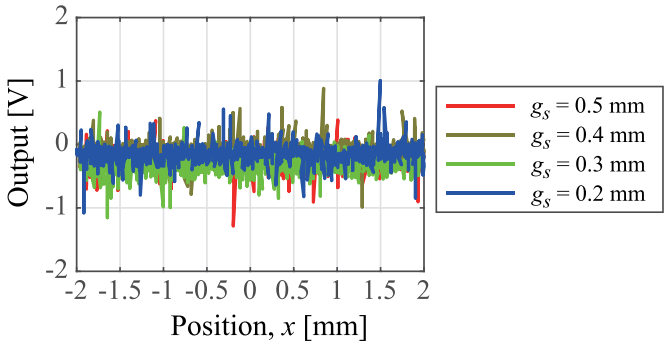


Figure 11. Measurement result for a slit thickness, t_s, of 0.3 mm and a slit opening, w_s, of 0.2 mm. No periodic signal can be seen in the plot, suggesting a failure of the measurement.

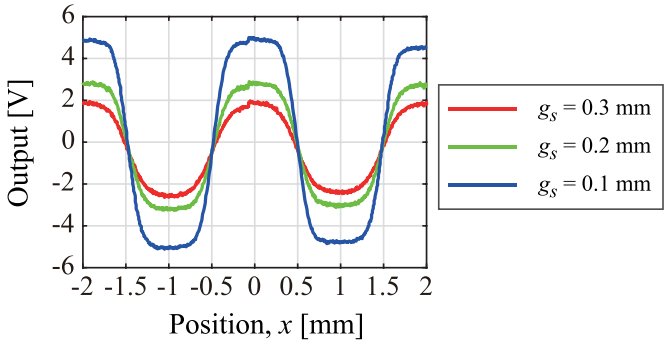


Figure 12. Measurement result for $t_s = 0.02$ mm and $w_s = 0.2$ mm. The slight discontinuities found around $x = 0$ are probably due to the deflection of a shim sheet by electrostatic attraction.

Figures 11 and 12 show the measurement results for the narrow slits ($w_s = 0.2$ mm). The result in figure 11 is for the thicker slit plate with the thickness of 0.3 mm. The measurement was repeated for different slit height, g_s, but the measured results did not show any periodic signal. This means that, as the simulation suggested, the electric field was almost completely shielded when a thick slit was used. On the other hand, the thinner slit plate ($t_s = 0.02$ mm) successfully realized measurement of the distribution, as shown in figure 12. The measured distributions look more like rectangular waves, which means that the slit realized the measurement of higher spatial frequency components; in other words, the spatial resolution was successfully improved.

In figure 12, due to the existence of the grounded plate, the measured voltage was considerably smaller than the real potential. This could be calibrated using the applied voltages to the target electrodes. For example, in figure 12, we know that the peak voltages are ± 100 V, which can be utilized for the calibration. Alternatively, we can measure a uniform potential distribution created by a single planar electrode to identify the sensitivity experimentally.

One technical concern found during the experiments is that, although it was not frequent, discharges happened between the grounded slit plate and the film substrate, especially when the height of the slit, g_s, was in a smaller range and the applied voltage was high. As the discharge damages the target film, it should be avoided. In the measurements of this paper, the gap and the voltage were adjusted not to cause discharges. If a higher voltage or a smaller gap is needed, one solution would be injecting dielectric liquid between the film and the slit plate.

3. Measurement of spatio-temporal distribution

The previous section showed that the measurement resolution can be improved by using a slit plate. In electrostatic devices, the charging of the surface insulating layers occur in various time ranges, depending on the conductivity of the insulating layer. To comprehend the performance of the electrostatic devices, visualizing such charging process is quite important. To meet such a demand, this section extends the proposed method such that it can visualize time variations of surface potential distributions. In electrostatic devices, the charging may occur in less than 1 s. Therefore, this work aims at visualization at a speed of more than 10 frames per second.

To visualize a temporal change of the distribution, the scanning as shown in figure 2 can be repeated at a high speed. In the original proposal in figure 2, either a set of the probe and the slit, or the target surface was moved by a linear motor stage. To realize a high speed scanning, however, the weight of the moving part should be minimized. Therefore, this section proposes to move the slit plate alone, as shown in figure 13. As the slit alone, which consists of shim tapes and an aluminium frame, is considerably light, high speed scanning motion can be easily achieved with a compact linear motor.

In the experimental setup, the aluminium frame of the slit was arranged on a linear guide such that it can move only in lateral direction. The frame was then connected to a voice coil motor and coil springs. The voice coil motor was driven with a sinusoidal voltage. Although the voice coil motor does not have a positioning function, the coil springs help to keep the position around their natural length. As a result, the slit plate oscillated around the natural length of the springs. During the oscillation, the position of the slit was monitored using a laser displacement sensor (LK-G85, Keyence) to identify the measured position.

During the oscillation of the slit plate, surface potential scanning can be achieved twice per one cycle of the oscillation. Therefore, the visualizing frequency for the potential distribution will be twice the driving frequency for the motor.

3.1. Effect of relative displacement of the slit

In the original method in figure 2, the relative position between the probe and the slit is fixed during the measurement. In the method proposed in this section, however, the relative position will change, which can affect the measured potential. Therefore, we first investigated the effect of the relative displacement and formulated a calibration function.

A surface with uniform potential was measured with different relative positions of the slit. Figure 14 shows the result, which was approximated using a Gaussian function expressed as

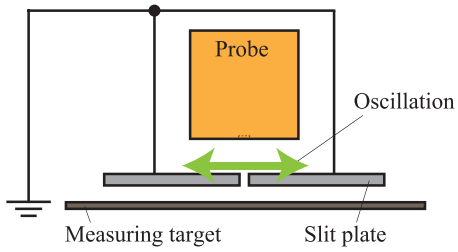


Figure 13. Proposed setup for spatio-temporal measurement. Only the slit plate is scanned to achieve high speed scanning motion. The probe and the target are fixed at constant locations.

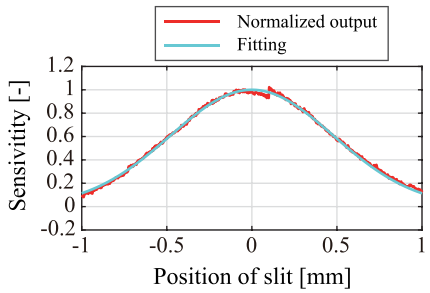


Figure 14. Normalized sensitivity against relative position of the slit. The slit used had a thickness of 0.02 mm and an opening of 0.2 mm. The sensitivity was maximized when the slit was located at the center and decreased as the slit shifted farther.

$$r(x) = \zeta \exp\left(-\frac{x^2}{2\sigma^2}\right) \tag{3}$$

where x is the relative displacement of the slit against the center of the probe. The parameters were identified as $\sigma = 0.48$ mm and $\zeta = 0.03$, for this particular case. Then, this function was used to compensate the measurement as,

$$v_{\text{comp}}(x) = \frac{v_{\text{raw}}(x)}{r(x)} \tag{4}$$

where v_{raw} and v_{comp} are the raw and compensated data.

3.2. Spatio-temporal measurement experiment

Using the proposed setup in figure 13, the film substrate in figure 8 was measured as in figure 15. The two-phase electrodes in the film were given a pair of 180-degrees phase-shifted sin voltages with a frequency of 2 Hz and an amplitude of 100 V. Then, the surface potential around one electrode was scanned. The estimated true temporal change of the surface potential is illustrated in figure 16.

The oscillation frequency of the slit, which is the frequency of the driving signal to the motor, was 20 Hz. This means the distribution was scanned 40 times per second. The slit had an opening of 0.2 mm and a thickness of 0.02 mm. The oscillation amplitude of the slit was about 0.6 mm. The scanned result is shown in figure 17. The results plotted are compensated using (4).

Comparing the two plots in figures 16 and 17, there are some differences in the details. However, the overall shapes are mostly in consistent, which verified the effectiveness of the proposed scanning method in terms of visualization of temporal changes. It is important to note that the frequency response of the surface voltmeter should be high enough to realize the visualization. Through the scanning, the spatial distribution is mapped to a temporal signal. If the frequency response of the voltmeter is limited, the measurement result would be filtered by the voltmeter's response. The surface voltmeter used in this work had a bandwidth wider than 100 Hz, according to our preliminary evaluation. If slower surface voltmeters are used, the oscillation frequency of the slit should be lowered.

Regarding the opening and the thickness of the slit, we expect that the same trends as in figures 4 and 7 apply to this spatio-temporal measurement. However, since the slit displaces from the center of the probe, a different trend may appear when the displacement is large. This point needs to be further investigated in future studies.

4. Application to an electrostatic film motor

It was reported that a synchronous electrostatic film motor can lose its output force when driven at a very low frequency [24]. As the motor is synchronous, the driving frequency is proportional to the speed of the motor. Therefore, when the motor moves at a very low speed or generates static thrust force at a constant position for a long time, the loss of output force can occur. The degree of the loss varies from motor to motor. For some motors, loss may happen under 0.1 Hz, whereas for other motors, it may happen under 0.01 Hz. The cause of the loss has not been fully clarified yet, but it is probably due to the charging of the insulation layer of the electrode films. Old and deteriorated films can be more easily charged and can cause the loss of force.

When the charging occurs on the insulating layer, the surface potential of an electrode film will decay. Therefore, by observing the temporal change of the surface potential under application of constant voltages, we can tell whether the films have good surface insulation or not. By doing so, we can estimate the motor's performance without assembling the motor, or we can exclude deteriorated films before assembling. In this section, we measure the surface potential of two different films, one is deteriorated and the other is not, and try to relate the result with the loss of the motor output force.

4.1. FPC film for electrostatic film motor

Figure 18 shows an FPC film used for an electrostatic film motor. The film has embedded parallel three-phase electrodes with an electrode pitch of 200 μm. The base layer is made of liquid crystal polymer (LCP), on which copper electrodes are patterned. Then, the electrodes are covered with an insulating layer, which is also made of LCP. The width of each electrode is around 70–80 μm. By applying three-phase voltages, the parallel three-phase electrodes create spatially periodic surface potential with a period of three electrode pitches, 600 μm.

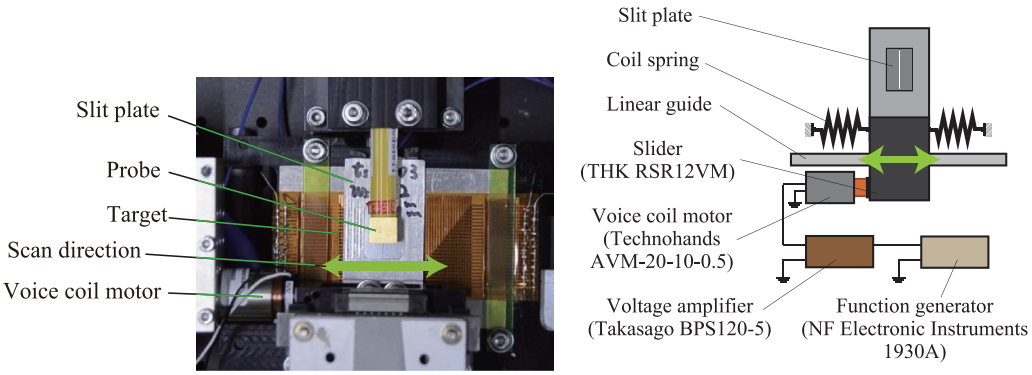


Figure 15. Experimental setup for spatio-temporal measurement with a scanning slit. The slit is oscillated in the lateral direction by a voice coil motor and a spring.

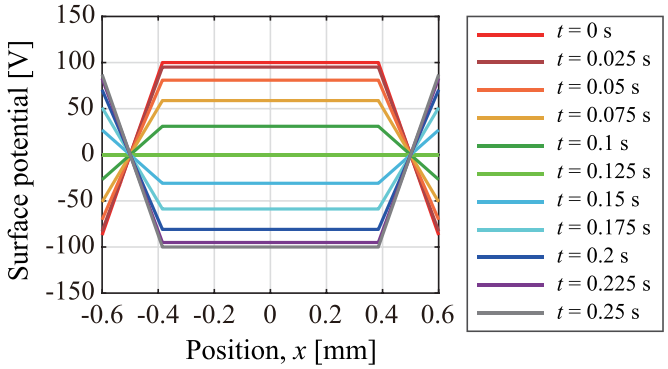


Figure 16. Estimated true potential distribution. The distribution oscillates at 2 Hz between 100 V and −100 V.

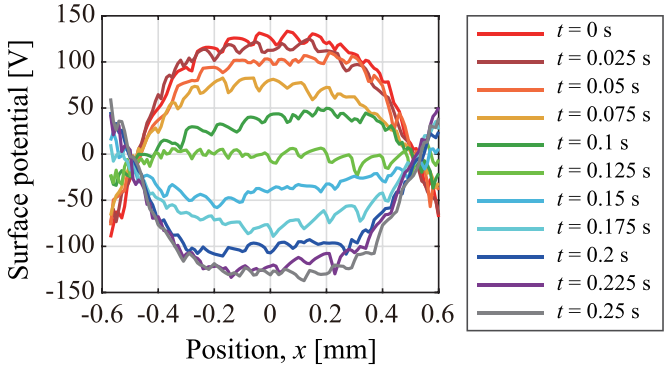


Figure 17. Measured spatio-temporal potential distribution. A slit with a thickness of 0.02 mm and an opening of 0.2 mm was actuated above the target surface.

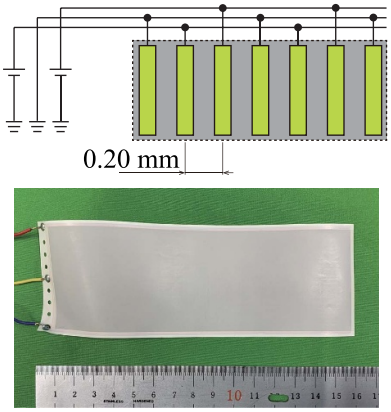


Figure 18. A film substrate for an electrostatic film motor. The film was fabricated as a flexible printed circuit. Parallel electrodes with a regular pitch of 0.2 mm are arranged on the base film. The width of each electrode is around 0.07–0.08 mm. The electrodes are covered with a cover film. The electrodes and the films are made of copper and liquid crystal polymer, respectively. The electrodes are connected in three-phase that generates a periodic potential pattern with a period of 0.6 mm when a set of three voltages is applied.

In an electrostatic motor, the film functions as a slider. To configure a motor, the slider will be combined with a stator film that also has three-phase electrodes with the same electrode pitch. For a synchronous operation at a constant speed, both of the slider and the stator electrodes are provided the same three-phase ac voltages. However, their connecting orders of the three-phase are reversed, such that the excited periodic potentials will travel to the opposite directions from each other. Between the two periodic potentials, electrostatic force will be generated to engage the two periodic potentials like gears. As a result, the slider will be propelled by the electrostatic force. The synchronous slider speed will be $6pf$, where p is the electrode pitch and f is the applied frequency [10].

4.2. *Surface potential measurement*

First, the surface of the slider film was scanned using the surface voltmeter without using a slit to confirm the insufficient resolution. The slider electrodes were given three-phase DC voltages of [+250 V, 0 V, −250 V]. Figure 19 shows the result of the surface potential measurement by scanning the surface with the probe of the voltmeter, without a slit. The output of the scanning showed almost nothing, which indicates that the spatial resolution is insufficient.

Next, the slider film was measured by the method proposed in the last section. In this measurement, three-phase DC voltages of [±100 V, 0 V, ∓100 V] were switched at a frequency of 0.2 Hz. Figure 20 shows the result, which was calibrated according to (4). The plot shows the time variation

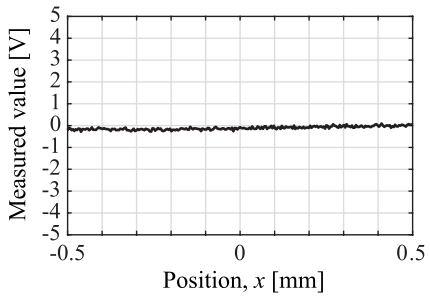


Figure 19. Measured surface potential distribution using the surface voltmeter alone. The surface of the film substrate with a periodic surface potential distribution was scanned with the probe of the voltmeter. Due to the insufficient spatial resolution of the voltmeter, the potential distribution was averaged to be zero.

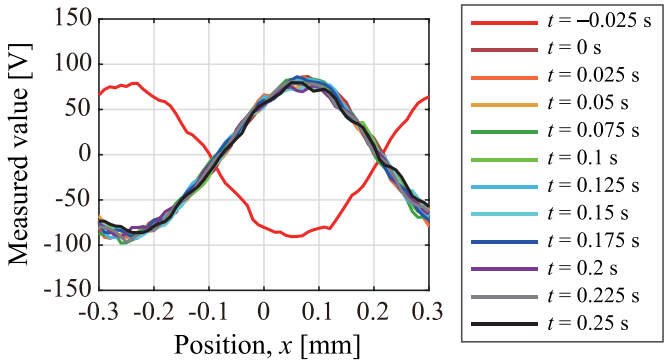


Figure 20. Time variation of the measured potential distribution for the film without surface treatment. Measured values were compensated with the predetermined calibration function. The voltages on the electrodes were switched at $t = 0$. The potential distribution was kept constant after the switching.

of the potential around a voltage switch at $t = 0$. It is clearly seen in the plot that the slider surface has a spatially periodic potential with a period of 0.6 mm. As the voltages on the electrodes were switched at $t = 0$, the spatial potential distribution was reversed, and then stayed in the same distribution after the switching. Although this plot describes only up to 0.25 s, we confirmed that the distribution did not change under the constant voltages for more than 30 s, at least in this voltage range.

To emulate a deteriorated film, another slider film with the identical design was prepared, which was then surface-treated using an anti-static spray (STATINOL S-250, Shishido Electrostatic, Ltd) to decrease its surface resistance. The film was given the same three-phase voltages, and its surface potential was measured. Figure 21 shows the result. Unlike the result in figure 20, the potential distribution decayed rapidly in figure 21. At $t = -0.025$, the previous distribution had already been decayed. Then, as the voltages were switched at $t = 0$, a new potential distribution appeared, but it decayed rapidly again. This means that the surface of the insulating layer was charged by the induced surface current.

Figure 22 shows how the potential decayed. The peak-to-peak value of the potential distribution at each time step was plotted over time. This plot shows that the potential of the treated film decayed with a time constant of about 0.1 s. If we

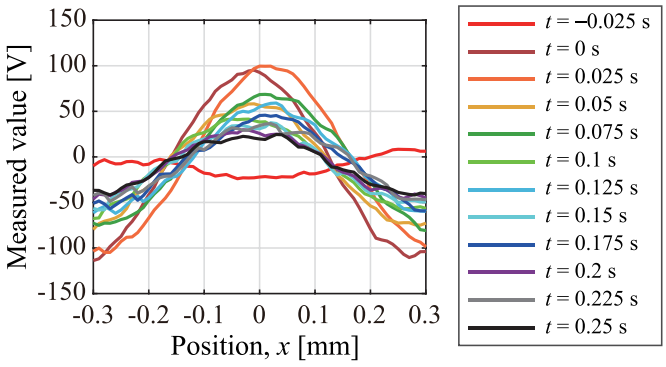


Figure 21. Time variation of the measured potential distribution for the film with surface treatment using an anti-static spray. Measured values were compensated with the predetermined calibration function. The measured potential decayed rapidly, after it appeared by the voltage switching at $t = 0$.

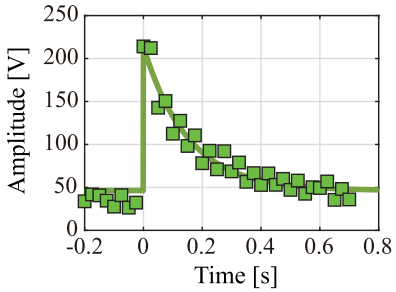


Figure 22. The change of the peak-to-peak value of the measured voltage distribution in the treated film. The potential decayed with a time constant of about 0.1 s.

approximate it by a single-order system, the time constant of 0.1 s corresponds to a break frequency of 10 rad s^{-1}.

4.3. Force measurement

Output thrust force of the electrostatic film motor was measured using the two slider films. In the setup shown in figure 23, a stator film was fixed to a base, on which a sliding part was arranged. To the sliding part, one of the two slider films was attached. Small glass beads with a diameter of 20 µm were scattered between the stator and the slider to maintain a gap. Then, the films were immersed in dielectric liquid (Fluorinert FC-770, 3M) to avoid discharge in air during high voltage application. For force measurement, the sliding part was connected to a load cell. The load cell was mounted on a micro stage, such that the sliding part can be moved. Since the motor has a force ripple, which means that the force characteristics has position dependency, the sliding part was manually moved to find the peak output force.

Figure 24 shows the output force measured at various frequencies. The phase amplitude of the applied three-phase voltage was 1 kV. The output decayed at lower frequency for both slider films. The reason for the loss of force for the non-treated film has not been clarified yet, but the difference of the two sliders probably came from the surface charging. Figure 25 shows the ratio of the output of the two films.

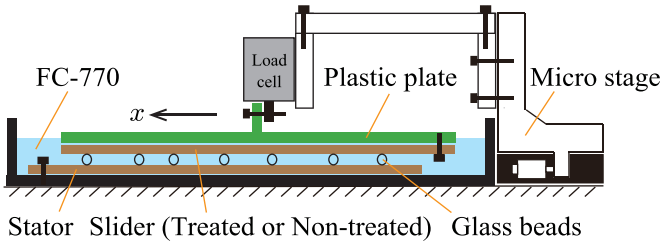


Figure 23. The experimental setup for measuring the output force of the motor. Two different sliders, one is surface-treated and the other is not, were utilized.

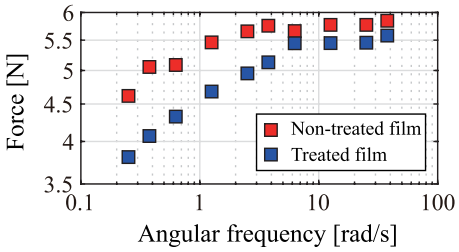


Figure 24. Measured peak output force for the two slider films. Surface treated film has a lower surface resistance and can easily be charged. It showed larger force loss at lower frequencies, compared to the non-treated film.

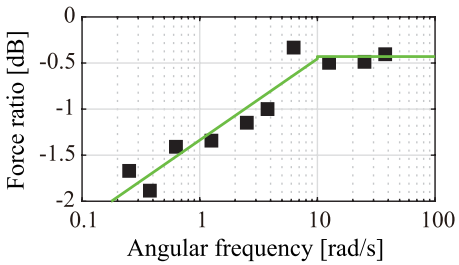


Figure 25. The ratio of the output force of the two films. The ratio decreased under a frequency of about 10 rad s^{-1}.

This shows that the loss of force started at around 10 rad s^{-1}, which coincides with the break frequency in the potential measurement.

It is not straightforward to relate the results of the potential measurement and the force measurement, since the electric fields in the two cases are different. During the force measurement, the slider film is placed over the stator film. As both of the films are provided voltages, the resulting electric field is considerably different from the field during the potential measurement. Also, the voltage ranges are different in the two measurements. The different voltages can affect the results, if the charging has voltage dependencies. Anther concern is that the surface coating using the spray was not very stable, and therefore there is a possibility that the surface conditions were different in the potential measurement and in the force measurement.

The force measurements showed that even the non-deteriorated film had the force losses at low frequencies, which cannot be explained by the results of the potential measurement. Although further investigations using a larger number of samples are required to reach a firm conclusion, the results showed a possibility that the motor performance can be evaluated through the surface potential measurement.

5. Conclusions

Fine parallel electrodes with an electrode pitch ranging from sub-mm to mm have been utilized in electrostatic devices. The electrodes create spatially-periodic surface potentials when multi-phase voltages are applied. The periodic potentials are then used to produce electrostatic force. Typically, the parallel electrodes are covered with surface insulating layers. If the insulating layers are charged, for example by surface current, the electric field from the electrodes will be altered that affect the performance of electrostatic devices. For example, in case of the electrostatic motor dealt in this work, the surface charging causes a loss of thrust force. To evaluate it, the surface potential distributions and their time variations need to be visualized. However, commercially available surface voltmeters cannot visualize it due to their limited resolution.

To address this issue, this work proposed a cost-efficient method to improve the resolution of surface voltmeters. The proposed method combines a thin slit plate to narrow the field of view. Through electrostatic field simulation and experiments, the work showed that the use of a thin-enough slit plate can successfully realize measurement with a fine spatial resolution.

Another advantage of using a slit is that by moving a lightweight slit at a high-speed, temporal change of surface potential distributions can be easily measured. The experimental setup demonstrated the measurement of surface potential distributions at 40 frames per second.

The proposed method was applied to an electrostatic film motor to show the effectiveness of the surface potential measurement in characterizing electrostatic devices. The potential measurement for a deteriorated electrode film showed that the surface charges up with a certain time constant, which coincided with the force characteristics of the assembled motor to some extent. Although the relevance of the force characteristics to the potential measurements needs to be further investigated, the results certainly showed the potential of the surface potential measurement in characterizing electrostatic devices.

In this paper, the cross-sectional shape of the slit was limited to a rectangle and the effects of the width and thickness were discussed. Different shapes, e.g. a slit with rounded edges, may change the characteristics, which might be an interesting topic for future research. Also, the field distortion caused by the slit plate should be studied in more detail. In the proposed method, a grounded slit plate is arranged close to the target surface, which could distort the electric field. The effect of the distortion needs to be investigated in future studies.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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References

[1] Yamamoto A, Nakashima T and Higuchi T 2007 Wall climbing mechanisms using electrostatic attraction generated by flexible electrodes *Proc. 2007 IEEE Int. Symp. on Micro-NanoMechatronics and Human Science* pp 389–94
[2] Prahlad H, Pelrine R, Stanford S, Marlow J and Kornbluh R 2008 Electroadhesive robots—wall climbing robots enabled by a novel, robust and electrically controllable adhesion technology *Proc. 2008 IEEE Int. Conf. on Robotics and Automation* pp 3028–33
[3] Liu R, Chen R, Shen H and Zhang R 2013 Wall climbing robot using electrostatic adhesion force generated by flexible interdigital electrodes *Int. J. Adv. Robot. Syst.* **10** 1–9
[4] Wang H, Yamamoto A and Higuchi T 2014 A crawler climbing robot integrating electroadhesion and electrostatic actuation *Int. J. Adv. Robot. Syst.* **11** 1–11
[5] Germann Jürg, Dommer M, Pericet-Camara R and Floreano D 2012 Active connection mechanism for soft modular robots *Adv. Robot.* **26** 785–98
[6] Chestney J A 1996 Dielectric selection for a robotic electrostatic gripping device *Proc. 7th Int. Conf. on Dielectric Materials, Measurements and Applications* pp 103–7
[7] Shintake J, Rosset S, Schubert B, Floreano D and Shea H 2016 Versatile soft grippers with intrinsic electroadhesion based on multifunctional polymer actuators *Adv. Mater.* **28** 231–8
[8] Guo J, Xiang C and Rossiter J 2018 A soft and shape-adaptive electroadhesive composite gripper with proprioceptive and exteroceptive capabilities *Mater. Des.* **156** 586–7
[9] Egawa S and Higuchi T 1990 Multi-layered electrostatic film actuator *Proc. I990 IEEE Workshop on Micro Electro Mechanical Systems* pp 166–71
[10] Niino T, Egawa S, Kimura H and Higuchi T 1994 Electrostatic artificial muscle: compact, high-power linear actuators with multiple-layer structures *Proc. 1994 IEEE Micro Electro Mechanical Systems Workshop* pp 130–5
[11] Nishijima T, Yamamoto A, Yasui H and Higuchi T 2006 A built-in displacement sensor for an electrostatic film motor *Meas. Sci. Technol.* **17** 2676–82
[12] Hosobata T, Yamamoto A and Higuchi T 2012 An electrostatic induction motor utilizing electrical resonance for torque enhancement *Sens. Actuators* A **173** 180–9
[13] Ge B and Ludois D C 2016 Dielectric liquids for enhanced field force in macro scale direct drive electrostatic actuators and rotating machinery *IEEE Trans. Dielectr. Electr. Insul.* **23** 1924–34
[14] Schaler E W, Zohdi T I and Fearing R S 2018 Thin-film repulsive-force electrostatic actuators *Sens. Actuators* A **270** 252–61
[15] Baoyun G, Ghule A N and Ludois D C 2019 High torque density macro-scale electrostatic rotating machines: electrical design, generalized *d-q* framework and demonstration *IEEE Trans. Ind. Appl.* **55** 1225–38
[16] Wang H, York P, Chen Y, Russo S, Ranzani T, Walsh C and Wood R J 2021 Biologically inspired electrostatic artificial muscles for insect-sized robots *Int. J. Robot. Res.* **40** 895–922
[17] Yamamoto A, Nagasawa S, Yamamoto H and Higuchi T 2006 Electrostatic tactile display with thin film slider and its application to tactile telepresentation systems *IEEE Trans. Vis. Comput. Graphics* **12** 168–77
[18] Bau O, Poupyrev I, Israr A and Harrison C 2010 TeslaTouch: electrovibration for touch surfaces *Proc. 23rd ACM Symp. on User Interface Software and Technology* pp 283–92
[19] Haga H, Yoshinaga K, Yanase J, Sugimoto D, Takatori K and Asada H 2015 Electrostatic tactile display using beat phenomenon for stimulus localization *IEICE Trans. Electron.* **E90-C** 1008–14
[20] Ilkhani G and Samur E 2018 Creating multi-touch haptic feedback on an electrostatic tactile display *Proc. IEEE Haptics Symp.* pp 163–8
[21] Isleyen A, Vardar Y and Basdogan C 2020 Tactile roughness perception of virtual gratings by electrovibration *IEEE Trans. Haptics* **13** 562–70
[22] Bulletti A, Capineri L, Materassi M and Dunn B D 2007 Surface resistivity characterization of new printed circuit board materials for use in spacecraft electronics *IEEE Trans. Electron. Packag. Manuf.* **30** 115–22
[23] Zavattoni L, Lesaint O and Gallot-Lavallée O 2013 Surface resistance measurements on epoxy composites: influence of surrounding gas, pressure, temperature and analysis of the origin of measured currents *2013 IEEE Int. Conf. on Solid Dielectrics (ICSD)* pp 370–3
[24] Niino T, Yamamoto A and Higuchi T 2000 Operation of a dual excitation multiphase electrostatic drive by amplitude-modulated ac voltage *Electr. Eng. Japan* **131** 78–84
[25] Vezzoli E, Amberg M, Giraud F and Lemaire-Semail B 2014 Electrovibration modeling analysis *EuroHaptics 2014: Haptics: Neuroscience, Devices, Modeling and Applications* (*Lecture Notes in Computer Science* vol 8619) (Berlin: Springer)
[26] Nakamura T and Yamamoto A 2017 Modeling and control of electroadhesion force in DC voltage *Robomech J.* **4** 18
[27] Bamber T, Guo J, Singh J, Bigharaz M, Petzing J, Bingham P A, Justham L, Penders J and Jackson M 2017 Visualization methods for understanding the dynamic electroadhesion phenomenon *J. Phys. D: Appl. Phys.* **50** 205304
[28] Guo J, Leng J and Rossiter J 2020 Electroadhesion technologies for robotics: a comprehensive review *IEEE Trans. Robot.* **36** 313–27
[29] Yahiaoui B, Megherbi M, Smaili A, Antoniu A, Tabti B and Dascalescu L 2013 Distribution of electric potential at the surface of corona-charged polypropylene nonwoven fabrics after neutralization *IEEE Trans. Ind. Appl.* **49** 1758–66
[30] Fatihou A, Dascalescu L, Zouzou N, Neagoe M-B Abdeljalil R and Laurentiu M D 2016 Measurement of surface potential of non-uniformly charged insulating materials using a non-contact electrostatic voltmeter *IEEE Trans. Dielectr. Electr. Insul.* **23** 2377–84
[31] Kikunaga K 2021 System for visualizing surface potential distribution to eliminate electrostatic charge *Sensors* **21** 4397
[32] Inoue T and Yokyama H 1994 Surface potential imaging of phase-separated LB monolayers by scanning Maxwell stress microscopy *Thin Solid Films* **243** 399–402
[33] Yasutake M 1995 Improvement of Kelvin probe force microscope (KFM) system *Jpn. J. Appl. Phys.* **34** 3403–5
[34] Uehara T, Higashio J, Ashizawa Y, Aizawa K, Nakagawa K and Itoh A 2009 Real time observation of surface potential distribution with an EFM on CTL *Proc. Int. Conf. on Digital Printing Technologies* pp 212–5

[35] Melitz W, Shen J, Kummel A C and Lee S 2011 Kelvin probe force microscopy and its application *Surf. Sci. Rep.* **66** 1–27
[36] Kawasaki T, Terashima T, Suzuki S and Takada T 1994 AC surface discharge on dielectric materials observed by advanced Pockels effect technique *J. Appl. Phys.* **76** 3724–9
[37] Kumada A, Sugihara T, Chiba M and Hidaka K 2002 Two-dimensional potential distribution measurement of surface discharge with subnanosecond resolution *Rev. Sci. Instrum.* **73** 1939–44
[38] Abolmasov S N, Abo R, Shirafuji T and Tachibana K 2006 Spatiotemporal surface charge measurement in two types of dielectric barrier discharges using Pockels effect *Jpn. J. Appl. Phys.* **45** 8256–9
[39] Yamano Y, Miyazaki M, Kobayashi S and Saito Y 2006 Measurement of 2-dimensional surface charge distributions under vacuum flashover events on insulators with sub-milli-second temporal resolution *Proc. Int. Symp. on Discharges and Electrical Insulation in Vacuum* pp 152–5
[40] Horenstein M N 1993 Measuring surface charge with a noncontacting voltmeter *Conf. Record of 1993 IEEE Industry Applications Conf.* pp 1811–6
[41] Noras M A and Pandey A 2010 Surface charge density measurements. Uses and limitations of Kevin probe-based instruments *IEEE Ind. Appl. Mag.* **16** 41–47